ConstruPilot

The deterministic construction SaaS — 0% hallucination

ALREADY LIVE · READY TO LAUNCH ACROSS EUROPE (EN / FR / PT)

Full dossier · foundations A to Z · Web Summit 2026 · Europe launch

Mission & founding principles

ConstruPilot is a SaaS platform for construction professionals and their clients. Its foundation is a simple promise: zero hallucination. All maths — takeoff, quotes, invoicing, energy and engineering — is deterministic, traceable and verifiable. AI is used only to read and assist; never to invent a number.

- 0% hallucination: all calculations are deterministic, never AI-generated.
- Full traceability: every data point carries its provenance (measurement / document / test) and status.
- AI reads (OCR, fact extraction) but never computes.
- Native regulatory compliance (Factur-X, energy/ADEME, building codes).
- No public claim without proof (claims ledger).
- Controlled operating budget with cost alerts.
- Multilingual and multi-VAT, ready for Europe (EN / FR / PT).

Technical foundations

- Frontend: React (CRA), shadcn/ui, Tailwind CSS; mobile apps via Capacitor (iOS + Android).
- Backend: FastAPI (Python) + MongoDB; rate limiting (SlowAPI).
- Security: HttpOnly cookies, JWT, Google sign-in, admin 2FA, brute-force protection.
- AI (read-only): Emergent universal key — OpenAI, Anthropic Claude, Google Gemini, Whisper (STT) and text-to-speech.
- Deterministic PDF generation (ReportLab); Factur-X PDF/A-3 + CII XML.
- Integrations: Stripe (payments), Autodesk APS (CAD), ADEME/France Rénov' (energy & grants), facturxapi (EN 16931), Perplexity (prices), ElevenLabs (voice), Resend (emails), Emergent object storage.

Modules, from A to Z

Module	Description
Construction AI assistant	Claude / OpenAI / Gemini (Emergent key): voice commands, web search for material prices, voice pricing — AI assists but never computes takeoff/pricing.
Plan takeoff	Imports PDF/DXF/images/ZIP, scale calibration, drawing of areas, lengths and volumes, opening deductions, families and subtotals, comparable versions.
Multi-level projects	Groups several plans into levels (Foundations, ground floor, 1st floor…) and consolidates takeoff + pricing per level and for the whole project, in a single PDF file.
Quote editor	Deterministic trade-by-trade pricing (16 works + custom items), editable unit prices, multi-rate VAT, discounts, deposit, payment schedule, professional PDF.
Factur-X invoices	Hybrid PDF/A-3 + CII XML e-invoices, BASIC and certified EN 16931 profiles, ready for mandatory e-invoicing.
Site tracking	Continuous site file: photos, reports, before/after comparator, handover minutes, reservation clearing — all timestamped and traceable.
Client space (portal)	Strictly isolated guest portal: the client follows the project, views quotes and invoices, accepts and signs online (OTP, automatic deposit).
RenoPilot — energy renovation	Imports official ADEME energy certificates, audit scenarios, grants (MaPrimeRénov'/CEE/eco-loan) via the France Rénov' API, remaining cost per action, PDF report.
StructuroPilot — structural engineering	Autonomous study from a description: fact extraction, adaptive questionnaire, traced model (levels/elements), sketches and a study note. Eurocode engine locked by design.
Document analysis	Reads Excel/PDF/scans with AI OCR, cell/page provenance, deterministic arithmetic checks, spec-to-quote reconciliation, correction memory.
Staircase calculation	Straight, quarter/half-turn and helical stairs compliant with building codes, optimization, concrete analysis, interactive 3D view, dimensioned PDF file, stair→quote.
Pro plan analysis	Processing of heavy CAD files (DWG/RVT/IFC) via Autodesk APS, for professionals.
Solve a problem	AI-assisted construction diagnosis from a description and photos (damp, cracks…).
HR & site time tracking	Geolocated clock-in/out, timesheets, selfie, site notes for field teams.
Company space & roles	Multi-user organization (owner, admin, site manager, accountant, member, worker), fine-grained permissions, KPIs, shared branding.
Admin & steering	Full back-office: user management, 2FA, Revenue Cockpit, expense tracking and a controlled operating budget.
Growth & public pages	Pro and consumer landing pages, narrated demo video, per-feature clips, SEO, public company showcase page.

The 3 mobile apps

ConstruPilot Business

Audience : Business owners, tradespeople, construction SMEs

The professional's field app: create quotes by voice, price on plans, track your sites and invoice without re-entering data.

- Compliant quotes & invoices (VAT, retention)
- Automatic takeoff on PDF plans
- Voice dictation: quotes, reservations, messages
- Continuous site file (photos, reports)
- Built-in client space

ConstruPilot Collaborator

Audience : Employees, journeymen and subcontractors

The field-team app: clock your hours, send your photos, flag reservations and check your daily tasks — everything flows back to the office.

- Geolocated clock-in/out (opt-in)
- Site photos and reports
- Reservation flagging
- Daily tasks

ConstruPilot Homeowners

Audience : Project owners / homeowners

Estimate the cost of your works in minutes, compare the quotes you receive and follow your project with full transparency.

- Costed estimate of works
- Quote comparison
- Site tracking (photos, reports)
- Access to renovation grants

Compliance & regulation

- Factur-X e-invoicing, certified EN 16931 profile (PDF/A-3 + CII XML).
- Official ADEME energy data (open-data certificates) and France Rénov' grants.
- Building-code-compliant calculations (staircases, concrete).
- GDPR: account and data deletion in-app and via dedicated URL.
- Apple/Google store compliance (privacy labels, Data Safety).

Market & Europe launch

ConstruPilot is operational and already live. The goal is a Europe-wide launch — starting with France and Portugal — powered by a multilingual (EN/FR/PT) and multi-VAT architecture, and an e-invoicing engine compliant with the European EN 16931 standard, ready for mandatory e-invoicing.

Contact: Martinho · ConstruPilot — ready for a live demo.